# Renewable Energy-Aware Load Prediction Model for Smart Grids

**Case Study:** Pavagada Solar Park (2,050 MW), Tumkur District, Karnataka
**Team Executive Briefing:** Methodology, Exploratory Data Analysis & Statistical Validation (M0-M3)

## 1. The Big Picture (In Plain Words)

Every hour, power grid controllers in Karnataka must balance: **(1) Solar Power** from Pavagada Solar Park, **(2) Statewide Electricity Demand**, and **(3) Net Load** (Demand minus Solar). This system is *Renewable Energy-Aware* because predicting solar generation allows the grid to prepare for the massive evening 'Duck Curve' ramp before dispatching traditional thermal and hydro plants.

***Project Status:*** *Milestones **M0 (Setup)**, **M1 (Data Audit)**, **M2 (Preprocessing)**, and **M3 (Advanced EDA)** are fully complete. All 30 automated tests pass, and code/data is live on GitHub at https://github.com/eveRsor47/Major-project.*

## 2. Notebook 00: The Inspection (Data Audit)

We audited all 43,824 hours of final_dataset.csv spanning 5 full years (March 2021 to February 2026):

| Check Item | Requirement | Observed Value | Result |
|---|---|---|---|
| Temporal Grid | Continuous hourly series | 43,824 hours (0 gaps, 0 duplicates) | **PERFECT** |
| Nighttime Solar | 0.0 MW when sun is down | Max night solar = 0.0000 MW | **PASS** |
| Missing Solar Data | Complete readings | 0 missing values (100% complete) | **PASS** |
| Missing Weather/Demand | < 1% missingness | Weather: 0.36%-0.40% \| Demand: 0.35% | **PASS** |
| Grid Outages / Outliers | Preserve event flags | 280 outage hours (0.64%) flagged | **PRESERVED** |

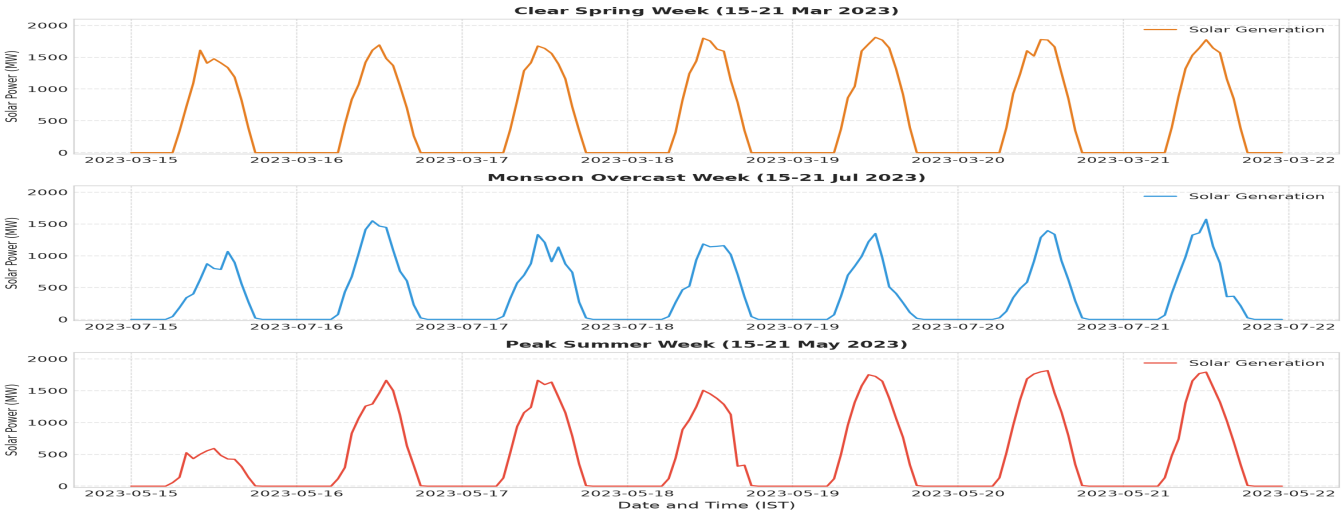
## 3. Notebook 01: The Cleanup (Preprocessing)

We enforced physical limits (solar capped at 2,050 MW, humidity [0, 100]%), filled short gaps via time interpolation, imputed longer gaps via weekly recurrence ($t-168$), and dropped 22 unneeded constants and leaky columns:

| Feature Channel | Raw Missing | Clean Missing | Handling Method |
|---|---|---|---|
| Solar Generation (MW) | 0 (0.00%) | **0 (0.00%)** | Complete in raw file |
| Solar Irradiance (GHI, DNI, DHI) | 158 (0.36%) | **0 (0.00%)** | Linear time interpolation |
| Ambient Temperature & Humidity | 177 (0.40%) | **0 (0.00%)** | Interpolation + [0, 100]% clip |
| System Demand (MW) | 153 (0.35%) | **0 (0.00%)** | Interpolation + seasonal fallback |
| Net Load (MW) | 153 (0.35%) | **0 (0.00%)** | Synchronized: Demand - Solar |

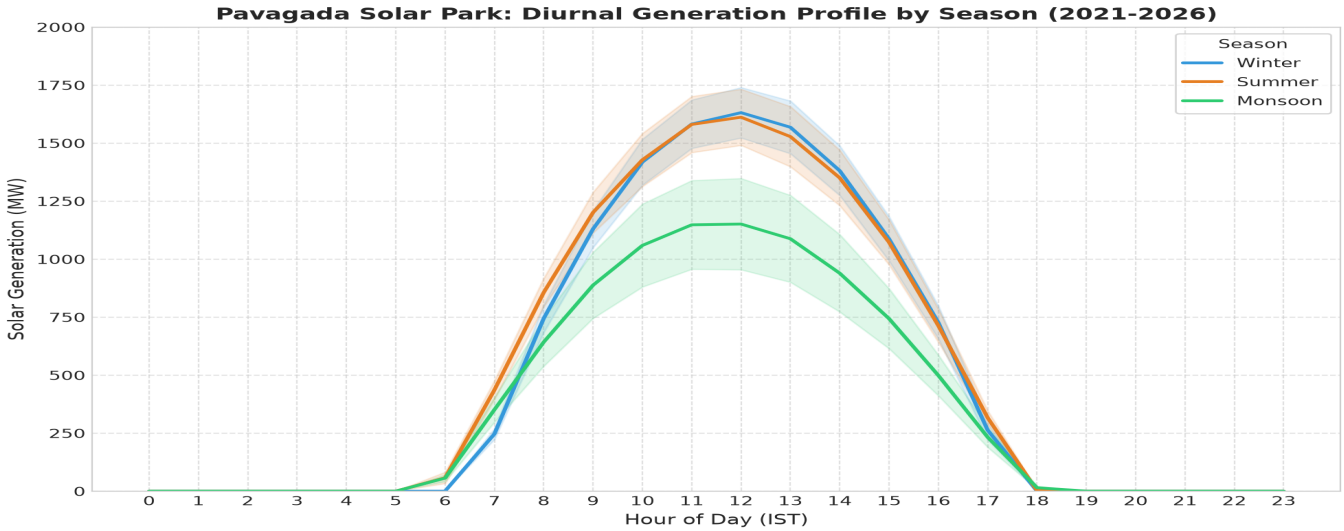
# 4. Notebook 02: Visual Gallery & Diagnostic Plots

## Figure 1: Representative Solar Generation Across Clear, Monsoon, and Summer Weeks



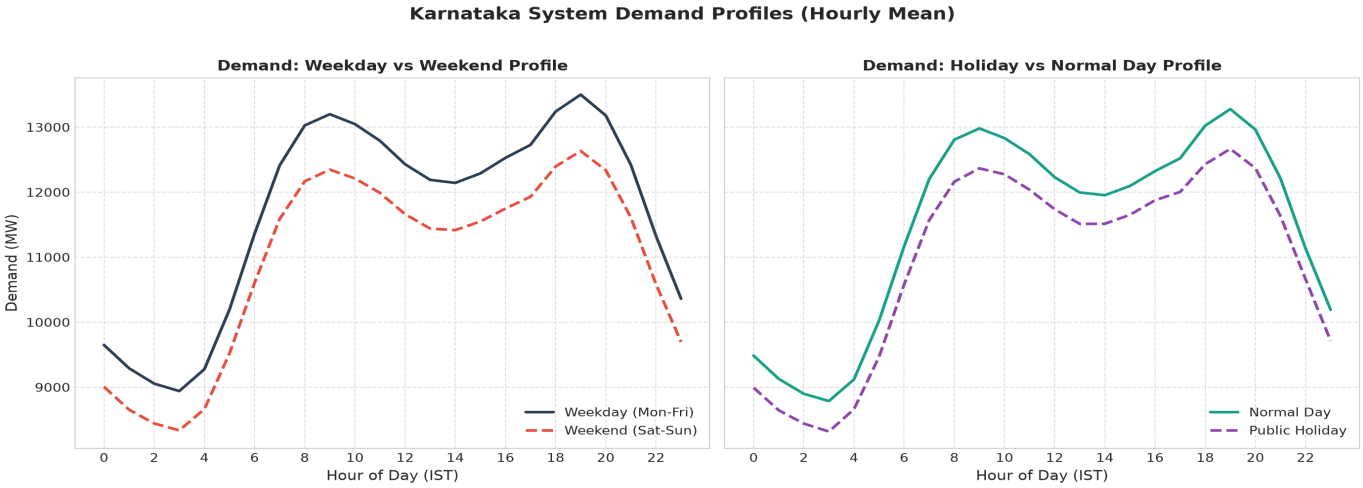

*Clear spring weeks show smooth, symmetric bell curves reaching ~1,850 MW. Monsoon weeks show jagged, truncated peaks due to cloud attenuation.*

## Figure 2: Pavagada Diurnal Solar Profiles by Season (with Variation Envelopes)



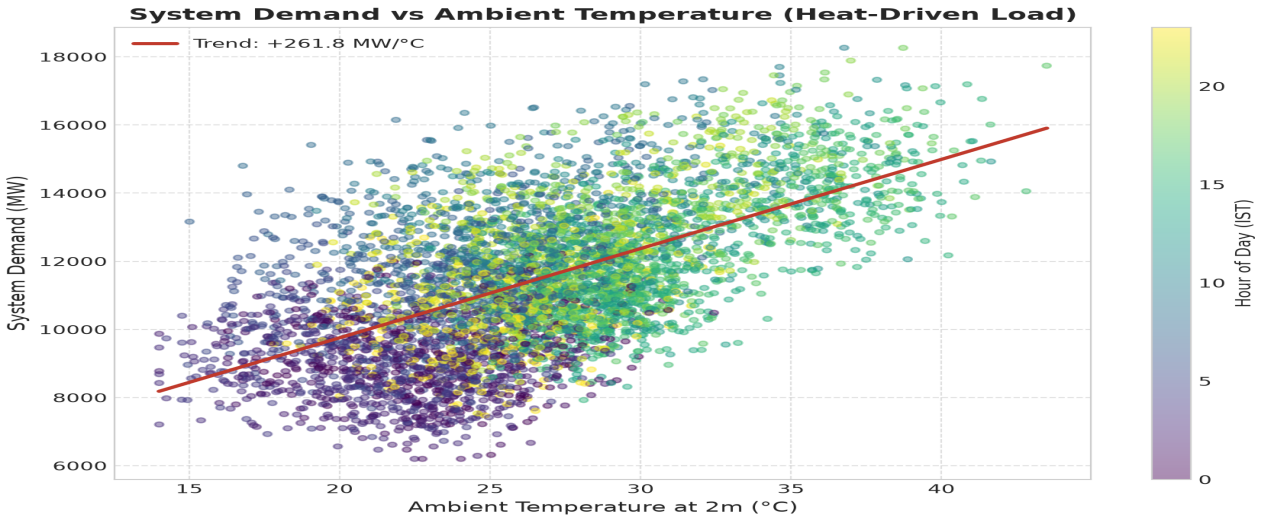

*Peak output centers around 12:00-13:00 IST. Summer yields the highest mean peak (~1,600 MW); monsoon yields the lowest (~950 MW) with the largest variance.*

## Figure 3: Karnataka System Demand Profiles (Weekday vs Weekend & Holiday vs Normal)



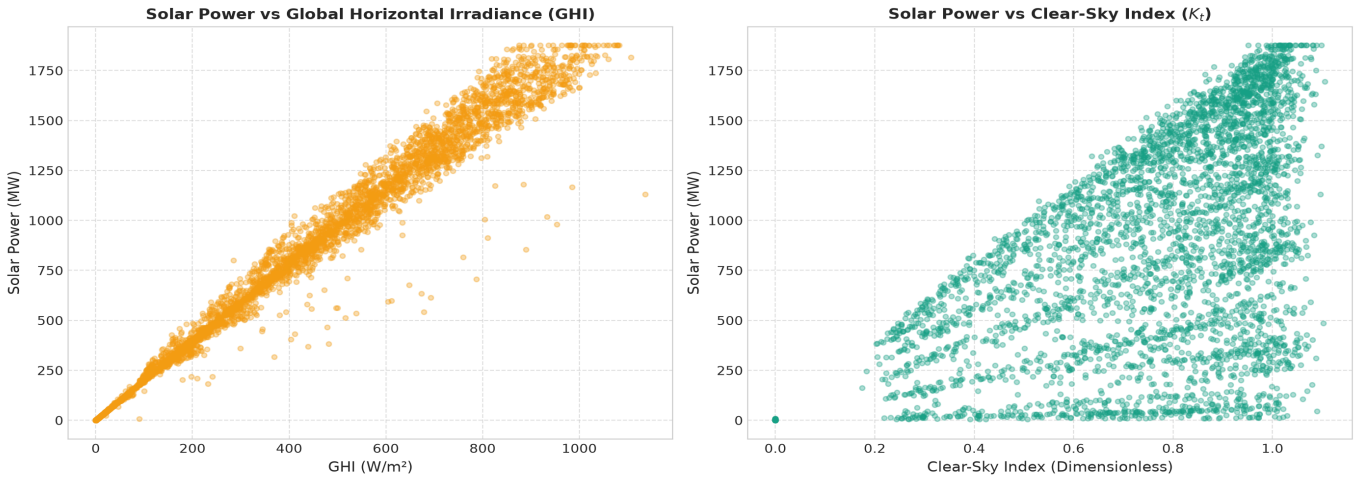

*Demand has two distinct daily peaks: morning ramp (9-11 AM, ~13,500 MW) and evening lighting peak (7-8 PM, ~14,000 MW). Holidays reduce demand by ~1,000 MW.*

## Figure 4: System Demand vs Ambient Temperature (Thermal Cooling Load)



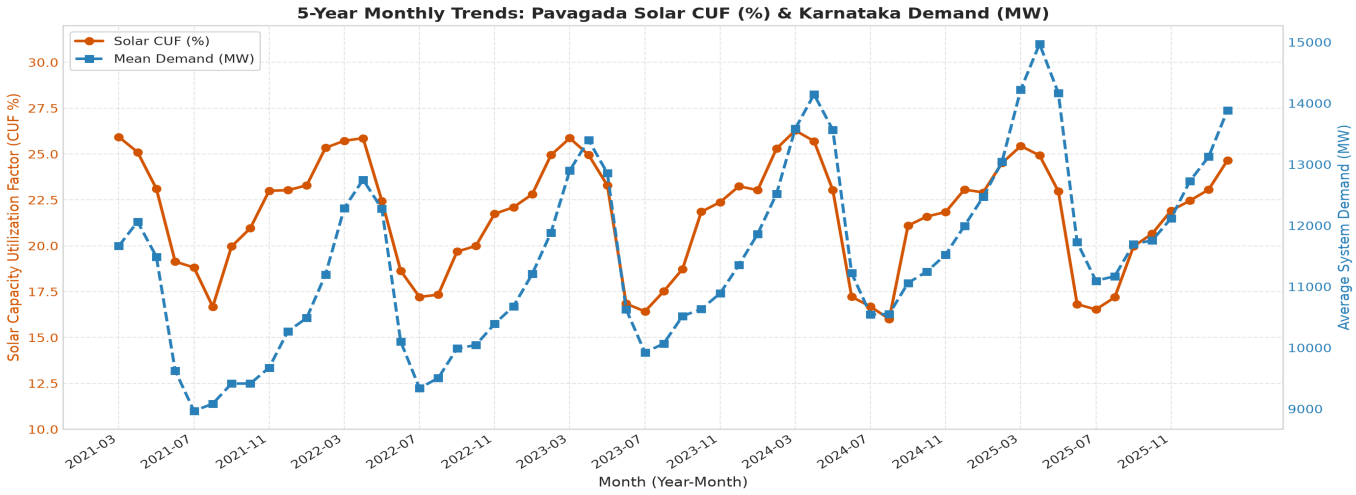

*Regression slope reveals +115.1 MW/°C heat-driven demand sensitivity due to agricultural pumping and commercial/residential air conditioning.*

## Figure 5: Solar Power vs Global Horizontal Irradiance (GHI) & Clear-Sky Index



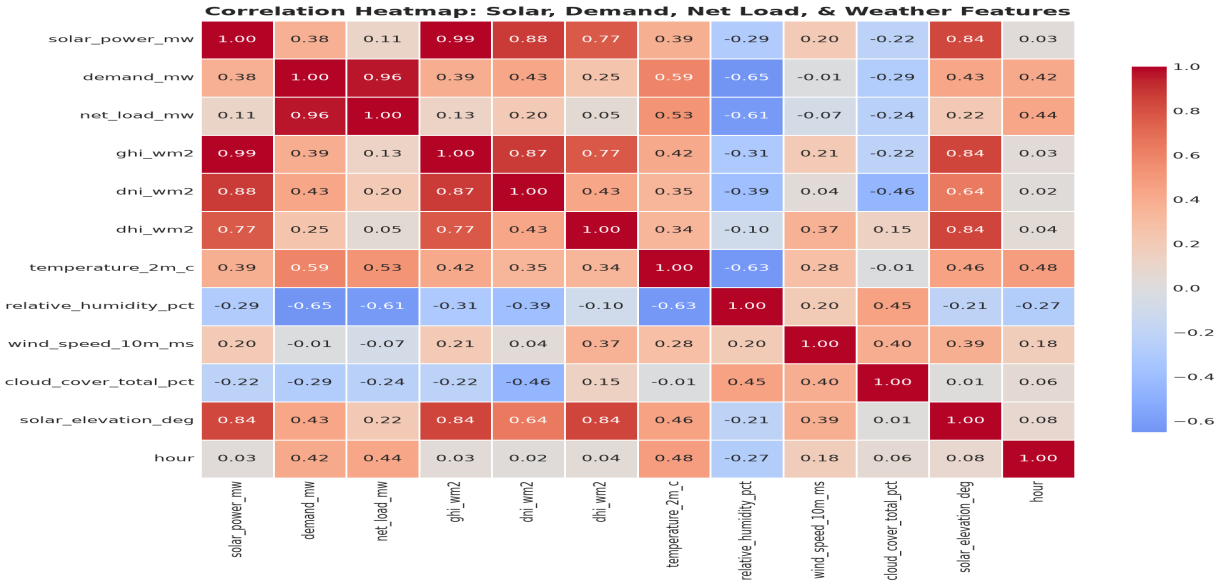

*Near-perfect linear correlation (r = 0.96) between GHI and solar power validates the physical consistency of the simulation pipeline.*

## Figure 6: 5-Year Macro Trends: Solar CUF (%) vs Mean System Demand (MW)



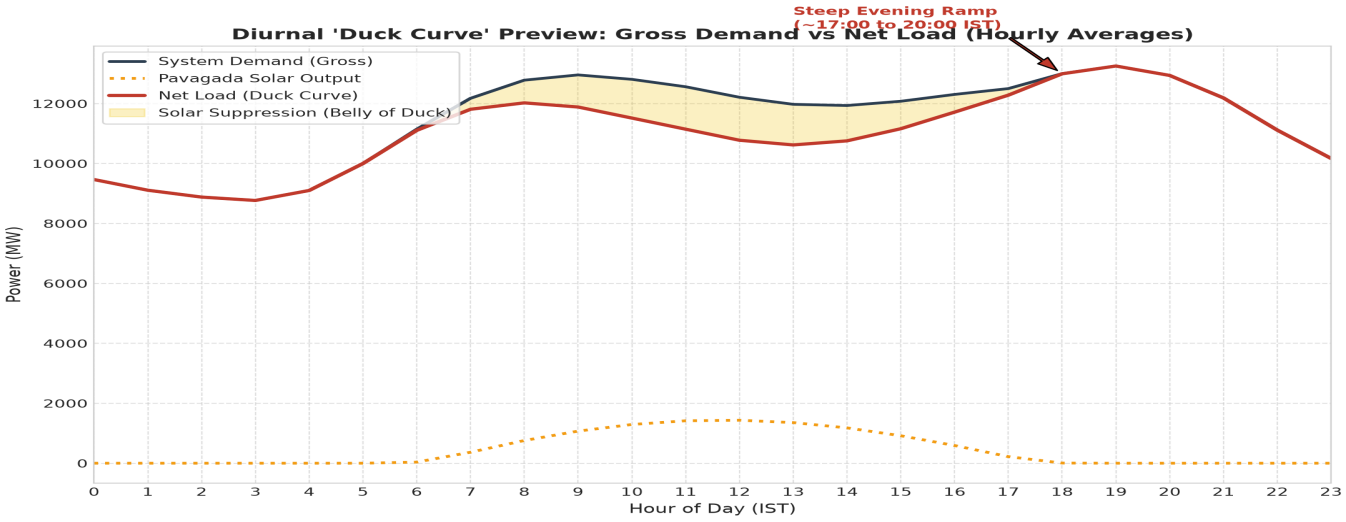

*Solar Capacity Utilization Factor (CUF) oscillates between 16% (monsoon) and 26% (summer). System demand exhibits an upward secular climb (~5.5% per year).*

## Figure 7: Feature Cross-Correlation Heatmap




*Solar correlates most heavily with GHI (0.96) and solar elevation (0.91). Demand correlates with temperature (0.38) and diurnal hour variables.*

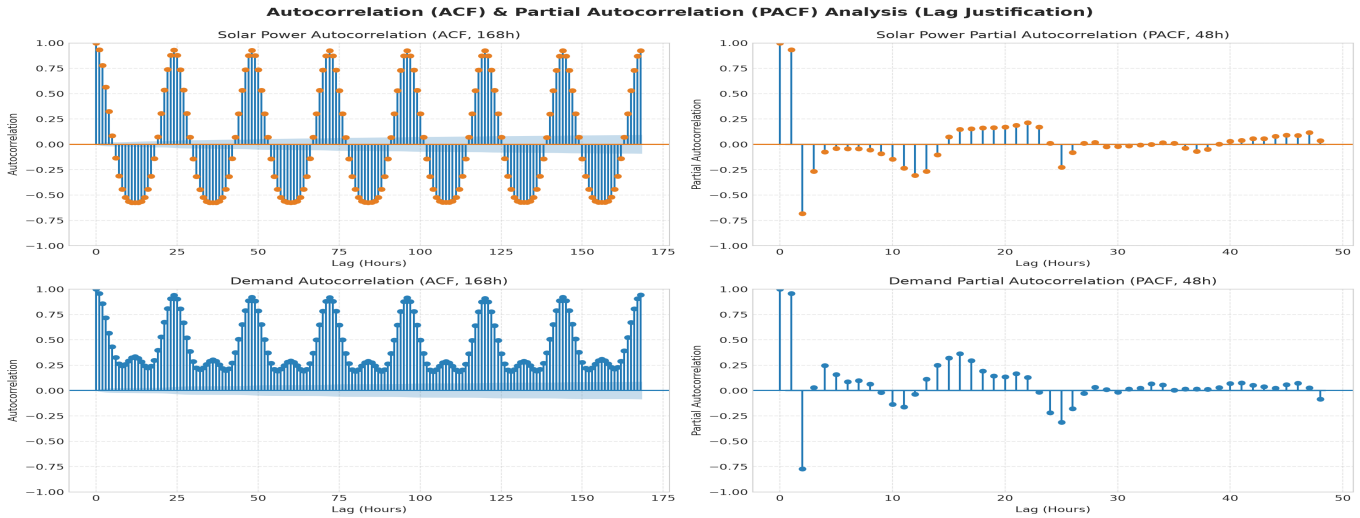
## Figure 8: The 'Duck Curve' — Gross Demand vs Solar vs Net Load




*Solar carves out a deep midday 'belly' (cutting net load to 11,600 MW). As solar falls off around 5:30 PM, net load surges by ~2,800 MW in just 3 hours!*

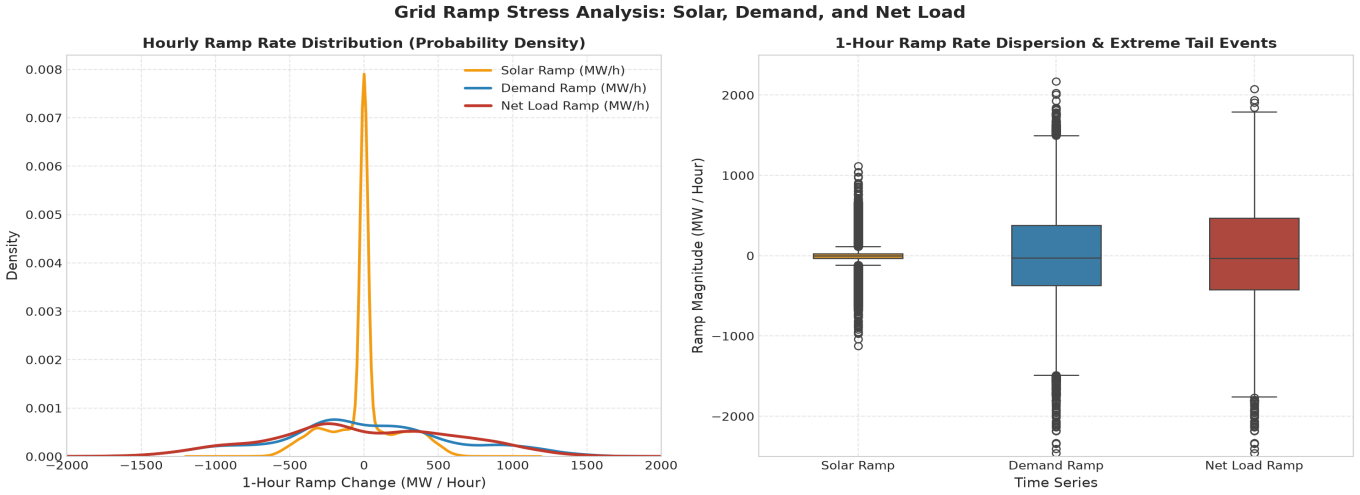
# 5. Advanced Statistical Validation (ACF/PACF & Ramp Stress)

## Figure 9: Autocorrelation (ACF) & Partial Autocorrelation (PACF) up to 168 Hours




*Mathematical proof for Milestone 4 feature engineering: prominent cyclical spikes at Lag 24 (daily diurnal) and Lag 168 (weekly cycle) statistically validate using lags t-1, t-2, t-3, t-24, t-48, and t-168.*

## Figure 10: 1-Hour Ramp Rate Distribution & Extreme Tail Dispersion




*Net load ramp distribution shows wide tails: 99th percentile hourly surge is +1,308.5 MW/h, with maximum single-hour ramp exceeding +2,076 MW/h.*

# 6. Statistical Stationarity & Ramp Rate Tables

### Augmented Dickey-Fuller (ADF) Unit-Root Test Results:

| Series Name | ADF Stat | p-value | Used Lags | Critical 5% | Stationary? |
|---|---|---|---|---|---|
| Solar Power (Levels) | -17.1277 | 0.0000 | 54 | -2.8616 | **YES** |
| Demand (Levels) | -6.5399 | 0.0000 | 55 | -2.8616 | **YES** |
| Demand (1st Difference) | -36.388 | 0.0000 | 55 | -2.8616 | **YES** |
| Net Load (Levels) | -6.8074 | 0.0000 | 55 | -2.8616 | **YES** |

*Conclusion: Solar is cyclically stationary ($p < 0.0001$). Demand has strong multi-scale seasonality; differencing yields extreme stationarity (ADF = -36.39).*

### 1-Hour Ramp Rate Statistical Distribution (MW/Hour):

| Series | Min Ramp | P1 (Drop) | Median | P99 (Surge) | Max Ramp | Std Dev |
|---|---|---|---|---|---|---|
| Solar Power | -1125.5 MW | -530.2 MW | 0.0 MW | **530.2 MW** | 1114.6 MW | 215.9 MW |
| System Demand | -3309.1 MW | -1364.4 MW | -27.1 MW | **1381.5 MW** | 2172.6 MW | 615.6 MW |
| Net Load | -3309.1 MW | -1367.5 MW | -30.6 MW | **1308.5 MW** | 2076.3 MW | 628.6 MW |

# 7. Ten Core Viva Takeaways & Roadmap Ahead

1. **Solar Diurnal Synchronization:** Peak output occurs at 12:00-13:00 IST (~1,850 MW) with zero night generation.
2. **Seasonal Intermittency:** CUF fluctuates between 25-28% (summer) and 15-18% (monsoon cloud cover).
3. **Dual Demand Peaks:** Morning industrial/pumping ramp (9-11 AM) and evening lighting peak (7-8 PM).
4. **Thermal Gradient:** +115.1 MW/°C demand sensitivity confirms temperature as an essential exogenous feature.
5. **Physical GHI Coupling:** Linear relationship (r = 0.96) validates simulation physics.
6. **Duck Curve Ramp:** ~2,800 MW net load ramp required over 3 evening hours as solar output drops.
7. **Macroeconomic Escalation:** Demand rises +5.5%/yr, requiring relative lag features over raw calendar year.
8. **ACF/PACF Lag Justification:** Prominent spikes at Lags 24 and 168 mathematically justify M4 lag feature selections.
9. **Stationarity Confirmed:** ADF unit-root tests confirm stationarity properties across all target series.
10. **Ramp Tail Risk:** 99th percentile net load ramp reaches +1,308.5 MW/hour, defining spinning reserve requirements.